Do Now · retrieval

Packet switching, circuit switching and standards

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Which TCP/IP layer contains TCP?

Q2

What does DNS usually provide when given a domain name?

Q3

Can a graph have more than one possible path between two vertices?

Q4

Name one advantage of sharing a network link instead of reserving it for one user.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Split a message into packets — The sender can split data into separate packets or segments. Each includes control information such as addressing and/or sequence information at relevant protocol layers.

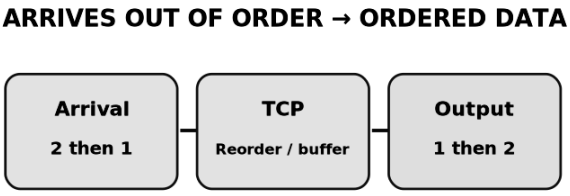
Core idea 2

Packets may take different routes — Packet switching lets network traffic share links. IP forwarding can choose different paths; it does not guarantee the same arrival order.

Core idea 3

Restore order when needed — TCP provides a reliable ordered byte stream, using acknowledgements and retransmission if required. Do not confuse TCP guarantees with best-effort IP delivery.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

packet switching — Data is split and forwarded across shared network links.

circuit switching — Communication uses a pre-established reserved path.

sequence information — Control information that allows ordered reconstruction.

retransmission — Sending missing or unacknowledged data again.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

Two TCP segments with sequence positions 1 and 2 travel on different routes and arrive 2, then 1. Explain the final data order.

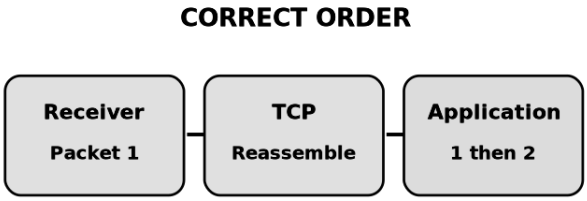
Follow the worked method

1. Route — The network may forward them along different paths, so arrival order is not guaranteed.

2. Receive — TCP uses sequence information to buffer out-of-order data.

3. Deliver — When missing earlier bytes arrive, TCP makes data available to the application in the correct order: first 1, then 2.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Detect gap
The receiver recognises that data between segment 1 and segment 3 is missing.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Give one difference between packet switching and circuit switching. [2]

Question 2

Data labelled 1,2,3 arrives as 2,1,3 using TCP. Explain how TCP handles the out-of-order arrival and what the application receives. [4]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Explain one advantage of packet switching and why communication standards are important for different manufacturers' devices. [4]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

IP guarantees that packets always arrive in the order sent.

Use key vocabulary from memory

1. Explain packet switching in your own words.

2. Explain circuit switching in your own words.

3. Explain sequence information in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.